Project Idea 2

Project Title

Hospital Medicine Delivery Robot

Purpose

Objective
Autonomously deliver medicines inside hospitals.

Scope
Hospitals, clinics, nursing homes.

Significance
Reduces staff workload and contact during infectious outbreaks.

Components

Microcontroller

- Arduino Mega

Sensors

- Ultrasonic Sensor
- IR Sensor
- RFID Reader

Actuators

- DC Motors
- Servo Motor
- Buzzer

Body

- Robot Chassis

Additional Components

- LCD Display
- Battery

Functionality

Functionality 1
Navigation using RFID checkpoints.

Functionality 2
Obstacle avoidance using ultrasonic sensor.

Functionality 3
Medicine compartment unlocks using servo after reaching destination.

Project Idea 6

Project Title

Autonomous Warehouse Inventory Robot

Purpose

Objective
Inspect warehouse shelves.

Scope
Warehouses.

Significance
Reduces inventory inspection time.

Components

Microcontroller

- Raspberry Pi

Sensors

- RFID Reader
- Ultrasonic Sensor
- Camera

Actuators

- DC Motors
- Servo

Functionality

1. Navigate aisles.
2. Scan RFID tags.
3. Capture inventory images.

Project Idea 9

Project Title

Autonomous Parking Assistant Robot

Purpose

Objective
Guide drivers toward available parking spaces.

Scope
Parking garages.

Significance
Reduces congestion.

Components

Microcontroller

- Arduino Mega

Sensors

- Ultrasonic Sensors
- IR Sensor
- RFID

Actuators

- Servo Barrier
- LED Indicators

Functionality

1. Detect empty spaces.
2. Open gate.
3. Display parking direction.

Project Idea 11

Project Title

Autonomous Smart Library Book Assistant Robot

Purpose

Objective
Transport books inside a library.

Scope
Libraries.

Significance
Reduces librarian workload.

Components

Microcontroller

- Arduino Mega

Sensors

- RFID
- Ultrasonic
- IR

Actuators

- DC Motors
- Servo

Functionality

1. Navigate shelves.
2. Deliver books.
3. Avoid obstacles.